Assembly Robot AR-6 Maintenance Manual

Six-axis assembly robot

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Norrland Robotics (fictional)

Read this manual before installing, operating or servicing the equipment. Keep it at the machine.

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1 Safety

The AR-6 is a six-axis assembly robot. Enter the cell only with the robot in manual mode with reduced speed (250 mm/s) and the enabling switch in your hand, or with the cell locked out.

2 Technical data

Item	Value
Payload	6 kg
Reach	900 mm
Axis motors	AC servo, 0.4–1.5 kW
Gearboxes	Strain-wave (axis 4–6), cycloidal (axis 1–3)
Supply	400 V, 16 A, 4.2 kW typical
Normal vibration RMS (axis 2 housing)	1.2–1.6 mm/s
Normal gearbox temperature	40–50 °C

3 Maintenance

3.1 Gearbox grease

Change the grease in axes 1–3 every 5,000 hours. Old, dark grease and metal particles point to reducer wear. Rising vibration and kurtosis on axis 2 usually mean wear of the reducer bearing (HDG-32-BRG).

3.2 Cabinet cooling

The controller cabinet fan (FAN-RB6) keeps the drives below 45 °C. A failed fan makes the axis motors and drives overheat within hours; vibration stays normal. Check the fan monthly.

3.3 Mounting and alignment

Loose gearbox mounting bolts on axis 3 cause a steady rise of vibration with normal kurtosis and position errors. Re-torque to 35 Nm and re-teach the reference position.

3.4 Cables

Inspect the cable harness on axes 4–6 every 6 months for wear at bending points.

4 Troubleshooting

Symptom	Cause	Action
Vibration up, kurtosis above 4	Reducer bearing wear	3.1
Vibration up, kurtosis normal, position error	Loose mounting	3.3
Motor hot, vibration normal	Cabinet fan failed	3.2
Collision message	Program, tooling	Check program